

Class #3 - Diagonalization



1. Theorem: For $A \in \mathbb{R}^{n \times n}$, if λ is an eigenvalue of A , then algebraic multiplicity of λ $\geq$ geometric multiplicity of λ .

Because eigenvectors corresponding to distinct eigenvalues are L.I.

If ~~an eigenvalue has~~ ^{Sum of} geo. mult. $>$ ^{Sum of} alg. mult.,
then $A \in \mathbb{R}^{n \times n}$ may have more than n _{of all eigenvalues}

L.I. eigenvectors.

But the max. number of L.I. vectors in $\mathbb{R}^n$ is n .

2. Definition: Let $A, B \in \mathbb{R}^{n \times n}$. The matrices A and B are called similar if there is an invertible $n \times n$ matrix P such that $A = P B P^{-1}$ (nonsingular)

$$(\Leftrightarrow P^{-1} A P = B)$$

Note that in such case, A and B have the same characteristic polynomial.

$$P_A(\lambda) = P_B(\lambda)$$

(proof)

Given A and B are similar, $A = P B P^{-1}$
for some $n \times n$ inv. matrix P .

$$\begin{aligned} P_A(\lambda) &= \det(A - \lambda I) \\ &= \det(P B P^{-1} - \lambda I) \\ &= \det(P (B - \lambda I) P^{-1}) \\ &= (\det P) (\det(B - \lambda I)) (\det P^{-1}) \\ &= (\det P) (\det(B - \lambda I)) \frac{1}{\det P} \\ &= \det(B - \lambda I) = P_B(\lambda) \end{aligned}$$

$P P^{-1} = I$
 $P I P^{-1} = I$ (scalar)
 $\lambda I = P (\lambda I) P^{-1}$
 A inv. $\det A^{-1} = \frac{1}{\det A}$
 $(\det A) (\det A^{-1}) = \det(A A^{-1}) = \det I = 1$

3. Theorem: Similar matrices have the same eigenvalues.

① Q #1 (10 min)

② 5 min. break

③ Attendance

④ Participation Quiz

⑤ Class #3 ~

⑥ Q #2 Next Mon. 9/21 (pp # 4, 5)

① $A = \begin{bmatrix} 2 & 4 & 3 \\ -4 & -6 & -3 \\ 3 & 3 & 1 \end{bmatrix}$

Char. poly $p(\lambda) = -(\lambda-1)(\lambda+2)^2$

② Eigenvalues λ_1, λ_2 in inc. order.

$\lambda_1 = -2, \lambda_2 = 1$

③ alg. multip. of λ_1 : 2
" of λ_2 : 1

④ $B = A - 1 \cdot I = \begin{bmatrix} 1 & 4 & 3 \\ -4 & -7 & -3 \\ 3 & 3 & 0 \end{bmatrix}$

$b_3 = -b_1 + b_2$ b_1, b_2 are L.I.

rank B = 2

nullity B = 1

geometric mult. of $\lambda = 1$ is 1

⑤ Eigenvector corresponding to 1.

$\begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix}$

$\begin{bmatrix} b_1 & b_2 & b_3 \end{bmatrix} \begin{bmatrix} 1 \\ -1 \\ 1 \end{bmatrix} = 0$

False

True

$\begin{bmatrix} 2 \\ -2 \\ 2 \end{bmatrix}$

can be an eigenvector corresponding to -2.

$A \vec{p}_i = \lambda_i \vec{p}_i$
 $A \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix} = -2 \begin{bmatrix} c_1 \\ c_2 \\ c_3 \end{bmatrix}$
 $\lambda_1 = -2$
eigenvector $\begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$

$\lambda_2 = 1$, eigenvector $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$
 $C = A + 2I$
 $= \begin{bmatrix} 4 & 4 & 3 \\ -4 & -4 & -3 \\ 3 & 3 & 3 \end{bmatrix}$
 $c_1 = c_2$
 $c_2 = c_1$
 c_1, c_2 are L.I.
rank C = 2
nullity C = 1
geo. mult. of $\lambda = -2$ is 1
L.I. $\begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$

Goal: Factorize $A = P D P^{-1}$ i.e., A is similar to a diagonal matrix.

$$\begin{bmatrix} 2 & 0 \\ 0 & -3 \end{bmatrix} \begin{bmatrix} 1 & 0 \\ 0 & \pi \end{bmatrix} = \begin{bmatrix} 2\pi & 0 \\ 0 & -3\pi \end{bmatrix}$$

4. Idea: Diagonal matrices are the easiest to deal with. (easy to multiply, easy to find eigenvalues, etc.)

Question: What kind of matrices are similar to a diagonal matrix?

Want A such that $A = P D P^{-1}$ for some P $n \times n$, inv.

$$AB = \begin{bmatrix} A \end{bmatrix} \begin{bmatrix} b_1 & b_2 & \dots & b_n \end{bmatrix} \Leftrightarrow A \begin{bmatrix} b_1 & \dots & b_n \end{bmatrix} = \begin{bmatrix} P D \end{bmatrix} \begin{bmatrix} b_1 & \dots & b_n \end{bmatrix}$$

$$= \begin{bmatrix} A b_1 & A b_2 & \dots & A b_n \end{bmatrix} \Leftrightarrow \begin{bmatrix} A p_1 & \dots & A p_n \end{bmatrix} = \begin{bmatrix} d_1 p_1 & \dots & d_n p_n \end{bmatrix}$$

$$\Leftrightarrow A \vec{p}_i = d_i \vec{p}_i \quad i=1, 2, \dots, n$$

$\vec{p}_i$ nonzero vector (P inv.)

Ex $A \in \mathbb{R}^{3 \times 3}$

$$AP = PD$$

$$\begin{bmatrix} A \end{bmatrix} \begin{bmatrix} \vec{x}_1 & \vec{x}_2 & \vec{x}_3 \end{bmatrix} = \begin{bmatrix} \vec{x}_1 & \vec{x}_2 & \vec{x}_3 \end{bmatrix} \begin{bmatrix} 2 & 0 & 0 \\ 0 & 4 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

Geo. mult. of $\lambda_1 = 2$: $\vec{x}_1, \vec{x}_2$ eigenvectors. $\uparrow$ L.I.

Geo. mult. of $\lambda_2 = 1$: $\vec{x}_3$ eigenvector. $\uparrow$ L.I.

5. Example: Let A be 5×5 real matrix with three distinct eigenvalues $\lambda_1, \lambda_2, \lambda_3$ with geometric multiplicities 2, 2, 1, respectively.

$$A = P D P^{-1} \quad ? \quad \text{Yes!}$$

If geometric multiplicities are 1, 2, 1?

$$A \neq P D P^{-1}$$

Not possible!

6. Def: An $n \times n$ matrix is called diagonalizable if it is similar to a diagonal matrix.
7. Theorem: An $n \times n$ matrix is diagonalizable $\Leftrightarrow$ the sum of the geometric multiplicities of all the eigenvalues equals to n .
8. Example: Diagonalize the following matrix, if possible.

$$\begin{bmatrix} 7 & 4 & 16 \\ 2 & 5 & 8 \\ -2 & -2 & -5 \end{bmatrix}$$

$$\det(A - \lambda I) = \begin{vmatrix} 7-\lambda & 4 & 16 \\ 2 & 5-\lambda & 8 \\ -2 & -2 & -5-\lambda \end{vmatrix}$$

$$p(\lambda) = -(\lambda-1)(\lambda-3)^2 = 0$$

This will be a Practice Prob.